SECTION A – 1 MARK QUESTIONS (10 × 1 = 10 Marks)

1. What is model drift?

a) Model loss increasing
b) Performance degradation over time as data distribution changes
c) Network latency
d) Storage issues

2. How is data drift detected?

a) Random sampling
b) Statistical tests comparing current vs. baseline distribution
c) Manual inspection
d) User feedback only

3. What is continuous monitoring in AI systems?

a) Running tests once
b) Real-time tracking of metrics, anomalies, and performance
c) Manual daily checks
d) Annual audits

4. Which metric is critical for chatbot deployment?

a) Model file size
b) Response latency, accuracy, user satisfaction
c) Training time
d) GPU memory

5. What is rate limiting in API deployment?

a) Slowing down the API
b) Restricting number of requests per user/minute to prevent abuse
c) Limiting API features
d) Blocking all traffic

6. How should AI system logs be managed?

a) Delete immediately
b) Store indefinitely
c) Structured logging with retention policy and access control
d) Never log

7. What is an AI alerting system?

a) Email notifications
b) Automated alerts when metrics exceed thresholds (performance drop, attacks)
c) Daily emails
d) Manual monitoring

8. What should a deployment checklist include?

a) Just code review
b) Security, performance, compliance, monitoring setup, documentation
c) Deployment only
d) Testing only

9. How is model performance tracked in production?

a) Manual spot checks
b) Continuous metrics dashboards, A/B testing, user feedback analysis
c) Once per month
d) Never

10. What is the purpose of incident response planning in AI systems?

a) Prevent all incidents
b) Have procedures ready for security breaches, model failures
c) Ignore failures
d) Blame users

SECTION B – 5 MARK QUESTIONS Answer ANY 6 (6 × 5 = 30 Marks)

1. Explain generative model types.

2. What is perplexity?

3. Define jailbreak attacks.

4. What is semantic search?

5. Explain AI content moderation.

6. Write short notes on Generative Adversarial Networks (GANs).

7. What is semantic similarity in AI?

SECTION C – 10 MARK QUESTIONS Answer ANY 4 (4 × 10 = 40 Marks)

1. Describe fine-tuning workflow step by step.

2. Explain AI risk assessment framework.

3. Discuss scaling generative AI systems.

4. Explain multimodal AI systems with examples.

5. Discuss AI bias and fairness in model training.

SECTION C – 20 MARK CASE STUDY Answer ANY 1 (1 × 20 = 20 Marks)

CASE STUDY 1: RECRUITMENT AI SCREENING SYSTEM

SCENARIO:

TalentFinder implements AI resume screening to process 10,000+ applications monthly. The system uses NLP embeddings to rank candidates. Issues:

AI systematically eliminates qualified candidates from underrepresented groups

Resume format and keywords bias against international candidates

Model trained on historical hiring perpetuates past discrimination

Lack of transparency in rejection decisions

No appeal mechanism for candidates wrongly filtered out

QUESTIONS

Q1 Conduct fairness audit identifying biases across demographic groups and propose data cleaning/rebalancing strategies.

Q2 Design an explainable AI system showing candidates why they were rejected with interpretable decision factors.

Q3 Develop appeal and human review mechanism ensuring no qualified candidates are unfairly eliminated by AI.

Q4 Create fairness constraints in model training ensuring equalized odds across protected characteristics (gender, ethnicity, age).

CASE STUDY 2: SOCIAL MEDIA CONTENT MODERATION

SCENARIO:

SocialConnect uses AI content moderation to filter harmful posts (hate speech, violence, misinformation). The system uses LLMs with safety filters. Challenges:

Over-moderation removes legitimate political discourse

Under-moderation allows hate speech in minority languages

Cultural context misunderstood: Sarcasm detected as harmful

No due process: Users don't understand why content was removed

Misinformation bypasses filters through encoded language

QUESTIONS

Q1 Design multi-language moderation framework ensuring quality across languages with cultural context awareness.

Q2 Implement confidence-based moderation system with human review for borderline cases and content appeals process.

Q3 Create guardrails preventing adversarial bypass techniques (encoded language, context manipulation) while allowing legitimate speech.

Q4 Develop transparency mechanism showing users moderation decisions with appeal process and statistical fairness reporting.